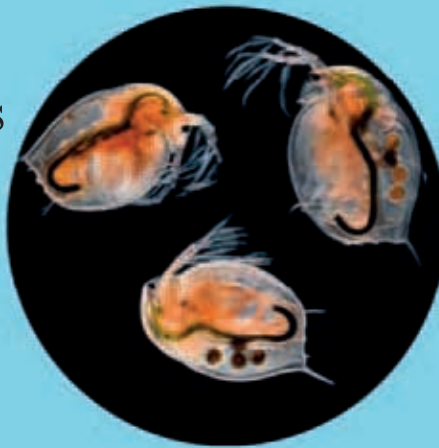


Swimming and diving

In oceans, rivers, lakes, and ponds, swimming animals are always on the move. To swim, they push against the water, often with flippers or fins. Fast swimmers need to be streamlined, since water is denser than air, but density also helps animals to float. Most swimming animals spend all their lives under water, but many air-breathing animals also dive down from the surface. Whales and seals dive while holding their breath for up to two hours at a time.

Mini swimmers

Ponds are full of miniature animals that swim to find food. These water fleas, or daphnia, swim by flicking their antennae, which look like feathery oars. Daphnia often swim toward the surface after dark to feed on algae and bacteria, and then sink again at dawn. Not much bigger than a pinhead, their top speed is about 0.01 mph (0.02 kph).



Jet-propellers

Trailing its legs behind it, this blue-ringed octopus uses jet propulsion to swim. It sucks water into a cavity in its body and then squirts it backward through a tube called a siphon. The force of the jet pushes the octopus in the opposite direction, making it swim headfirst. This way of swimming is used by other related mollusks, including nautiluses and squid, and also by dragonfly larvae in freshwater.



Diving deep

Elephant seals can move quickly on land, but they are even more impressive under water. They eat squid, octopuses, and deep-sea fish, sometimes diving more than 3,300 ft (1,000 m) to find their food. They can stay under water for up to two hours. Elephant seals manage this incredible feat by storing oxygen in their muscles and blood while they are resting at the surface. They breathe out just before they dive, so their lungs contain only a small amount of air.



Specialized swimming



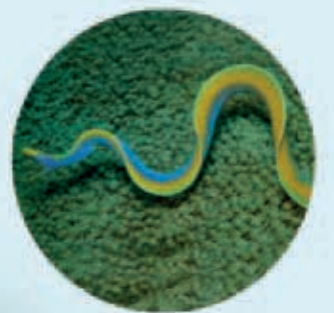
Rays

A ray's broad pectoral fins look like a pair of wings. Rays that are bottom-feeders swim by rippling their fins, but those that live in open water, such as this giant manta ray, usually swim by beating theirs up and down. These rays have the largest "wingspan," which can measure more than 23 ft (7 m) across.



Sea horses

Unlike most fish, seahorses swim with their bodies upright. They push themselves along by fluttering a small fin on their backs and steer by using two tiny fins on either side of their heads. When they are not swimming, they use their tails to hook onto corals and seaweeds.



Eels

Many adult eels, such as this ribbon eel, have a single narrow fin along the length of their bodies. They swim by curving their bodies in the same kind of movement used by swimming snakes. Most eels spend all their lives in water, but some also crawl out and wriggle over land.